

Ryan A. Marsala

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Expecting to start a PhD in computer science August 2026. Primarily focused on HPC, simulation and embedded programming.

EDUCATION

M.S. Computer Science

GPA: 3.83

Colorado School of Mines, Golden, CO

Aug 2023 – Aug 2025

Thesis: *End-to-end Simulation Framework for Hybrid SFQ/CMOS-Memory Compute Systems*

B.S. Double Major: Electrical Engineering & Computer Science

RESEARCH

Project: Modeling the Memory-Compute Gap in Large-scale Superconductive Systems

Aug 2023 – Aug 2025

Collaborative Project with Rochester and UVA professors and students.

- **Paper accepted for ISPASS 2026.** Poster accepted and presented at OSCAR 2025
- **Objective:** To build a novel simulation framework for identifying performance bottlenecks in next-generation superconducting computer architectures, modeling unique cryogenic characteristics
- Explored modeling and analyzing novel superconducting computers utilizing RSFQ logic
- Developed analytical models to characterize various physical characteristics of cryostats and their interconnects
- Integrated cache modeling and latency characterization specific to cryogenic environments
- Implemented cache simulation, RISC-V instruction decoding, and synthesized RISC-V traces

WORK EXPERIENCE

Research Intern, Lawrence Berkeley National Lab

June 2025 – Oct 2025

Collaborative Project involving AMCR, CAD, computer science, and physics division scientists

- **Objective:** Exploring noise modeling and analysis on superconducting ADCs by developing MATLAB and simulink models to validate their performance and pinpoint critical error sources
- Created noise models for characterizing superconducting specific intrinsic noise effects such as pulse jitter
- Improved on ADC simulink designs for improved accuracy, parallel simulation and 3x runtime improvement
- Improved SNDR calculations by 100% for spectral analysis using techniques such as Blackman-Harris windowing
- Explored mitigation methods for noise sensitive areas and their effects on improving design robustness
- Presented methodology and results at CSASP summer intern poster session

Canary X Sensor, Lunar Outpost

Field session project commissioned and managed by Lunar Outpost engineers

- **Objective:** Develop a micro controller solution for periodically communicating sensor information to on-site servers
- Designed custom firmwares for encryption and dynamic transmission through ethernet, wifi, and cellular networks
- Implemented various inter-device communications and logic using UART, SPI and I2C connections
- Made specialized data retention and state machines for robust logic and fault tolerance with 99.9% uptime
- Awarded 1st place in technical presentation of the final implementation

PUBLICATIONS

Marsala, R. A., et al. "Hibiscus: End-to-end Architectural Simulation Framework for Hybrid SFQ/CMOS-Memory Compute Systems". Poster presented at OSCAR 2025. **Accepted for ISPASS 2026**

Marsala, R. A., et al. "Noise Analysis on Superconducting ADCs". In submission for NEWCAS 2026

INTERESTS AND PROJECTS

- Member of bowling club and rhythm club, involved in hosting various events
- Various GPU, parallel, web, android and game programming projects including CUDA, OpenMP and OpenGL
- Various electrical projects including an autonomously navigating robot, PCB design, BPM detector, and FPGA boards

SKILLS & STRENGTHS

Programming: C++, C, Java, Python, JavaScript, PHP, Kotlin, MATLAB, LabVIEW, SQL, Verilog, Lisp, HPC, PCB

API/ABIs: OpenGL, GLFW, GLSL, CUDA, OpenMP, OpenCV, RISC-V, Android, Next.js, PyTorch

Electrical Experience: Microcontrollers, Motors and Generators, PCB design, Soldering, CAD

Software: Git, Autodesk EAGLE, Linux, Microsoft Office, SolidWorks, VSCode, Docker, Simulink, Traces, WRspice

Professional Skills: Documentation, Collaboration, Communication, Adaptability, Problem Solving, Teamwork